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The Effects of Diffusion Coefficients in a Two-Species Lotka-Volterra Competition System with Resource Dependent Dispersal

Qi Wang[1]

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Abstract
In this paper, we consider a Lotka-Volterra competition-diffusion system with resource-dependent dispersal. We study the linear v.s. global asymptotic stability of steady states. Furthermore, how the diffusion coefficients and the dispersal strategies of two competing species affect the stability of steady states are given. This paper is a further study of (Tang and Wang in J. Math. Biol. 86:23, 2023).

Keywords Resource dependent dispersal · Coexistence · Asymptotic stability · Reaction-diffusion equations

Mathematics Subject Classification 35K57 · 35K61 · 37C65 · 92D25

1 Introduction

In the past couple of decades, reaction-diffusion equations have been frequently used to model the dispersal process and its ecological effects in population dynamics, and are widely used to describe dispersal interm of diffusion. One classical example of these models is a two-species Lotka-Volterra competition-diffusion system as follows; see, e.g., the book [2].

$$\begin{cases} u_t = \mu_1 \Delta u + u(r_1(x) - u - cv), & \text{in } \Omega \times (0,T), \\ v_t = \mu_2 \Delta v + v(r_2(x) - bu - v), & \text{in } \Omega \times (0,T), \\ \dfrac{\partial u}{\partial n} = \dfrac{\partial v}{\partial n} = 0, & \text{on } \partial\Omega \times (0,T), \\ u(x,0) = u_0(x) \geq, \not\equiv 0, & \text{in } \Omega, \\ v(x,0) = v_0(x) \geq, \not\equiv 0, & \text{in } \Omega. \end{cases} \tag{1.1}$$

Here $u(x,t)$ and $v(x,t)$ represent the population density of two competing species at location $x \in \Omega$ and time $t > 0$, respectively; Ω, the habitat, is a bounded smooth domain in

✉ Q. Wang
qwang@usst.edu.cn

[1] College of Science, University of Shanghai for Science and Technology, Shanghai 200093, P.R. China

$\mathbb{R}^n$, n denotes the outward unit normal vector on the boundary $\partial\Omega$; $\mu_1, \mu_2 > 0$ are random diffusion rates of species u and v, respectively; $b, c > 0$ measure the inter-specific competition ability, while the intra-specific competition coefficients are normalized by 1 here; $r_1(x), r_2(x) > 0$ stand for the environmental resource shared by species u and v, respectively, which sometimes are also called environment/resource functions since they can reflect the environmental situations.

To our knowledge, currently there have been many works attempting to understand the population dynamics of system (1.1); see, e.g., [5–7, 15] and the references therein. As a further development of model (1.1), recently there is a interest in the study of population dynamics of the following two species competition diffusion model with resource-dependent dispersal in [17].

$$\begin{cases} \partial_t u_1 = \mu_1 \Delta(d_1(r)u_1) + u_1(r(x) - u_1 - u_2), & \text{in } \Omega \times (0, T), \\ \partial_t u_2 = \mu_2 \Delta(d_2(r)u_2) + u_2(r(x) - u_1 - u_2), & \text{in } \Omega \times (0, T), \\ \dfrac{\partial(d_1(r)u_1)}{\partial n} = \dfrac{\partial(d_2(r)u_2)}{\partial n} = 0, & \text{on } \partial\Omega \times (0, T), \\ u_1(x, 0) = u_{1,0}(x) \geq, \not\equiv 0, & \text{in } \Omega, \\ u_2(x, 0) = u_{2,0}(x) \geq, \not\equiv 0, & \text{in } \Omega. \end{cases} \tag{1.2}$$

Compared to the model (1.1), some influences of the resource dependent dispersal rates on the local v.s. global dynamics of populations were studied in [17]. However, as far as we know, the classification of global dynamics was not fully described yet. Motivated from [17], in this paper, we will address this issue about system (1.2) by further specifying different environmental resource, i.e., we shall consider the following model.

$$\begin{cases} \partial_t u_1 = \mu_1 \Delta(d_1(r_1)u_1) + u_1(r_1(x) - u_1 - u_2), & \text{in } \Omega \times (0, T), \\ \partial_t u_2 = \mu_2 \Delta(d_2(r_2)u_2) + u_2(r_2(x) - u_1 - u_2), & \text{in } \Omega \times (0, T), \\ \dfrac{\partial(d_1(r_1)u_1)}{\partial n} = \dfrac{\partial(d_2(r_2)u_2)}{\partial n} = 0, & \text{on } \partial\Omega \times (0, T), \\ u_1(x, 0) = u_{1,0}(x) \geq, \not\equiv 0, & \text{in } \Omega, \\ u_2(x, 0) = u_{2,0}(x) \geq, \not\equiv 0, & \text{in } \Omega. \end{cases} \tag{P}$$

1.1 Development and Related Work

In the past few years, various variants of system (1.1) have been qualitatively investigated. When both the resources is spatially homogeneous, namely both r_1 and r_2 are positive constant, the global dynamics of (1.1) crucially depends on the ecological reaction coefficients (e.g. see [3]). More precisely, the positive coexistence equilibrium $(\frac{r_1 - cr_2}{1 - bc}, \frac{r_2 - br_1}{1 - bc})$ is globally asymptotically stable if $0 < b < \frac{r_2}{r_1} < \frac{1}{c}$; while competitive exclusion equilibrium $(r_1, 0)$ (resp. $(0, r_2)$) is globally asymptotically stable if $\min\{b, \frac{1}{c}\} \geq \frac{r_2}{r_1} > 0$, $(b, c) \neq (\frac{r_2}{r_1}, \frac{r_1}{r_2})$ (resp. $0 < \max\{b, \frac{1}{c}\} \leq \frac{r_2}{r_1}$, $(b, c) \neq (\frac{r_2}{r_1}, \frac{r_1}{r_2})$).

When at least one of r_1 and r_2 is nonconstant, system (1.1) is much more harder to deal with. He and Ni gave some novel results about the dynamics of (1.1) in [5]. The basic as-

sumption in [5] on parameters b, c includes the range $\{(b,c): b,c>0, bc\le 1\}$, which covers the weak competition case $\{(b,c): 0<b,c<1\}$ that was previously discussed by Lou in [11] as well as the special case of identical competition ability $b=c=1$ that was studied by Dockery et al.in [4]. The resource functions $r_1(x)$ and $r_2(x)$ are supposed to enjoy the following conditions: $r_i(x)\in C^{\alpha}(\Omega)(\alpha\in(0,1))$ with $\int_\Omega r_i(x)\mathrm{d}x\ge 0$ and $r_i(x)\not\equiv 0$, $i=1,2$. Under the above conditions, He and Ni [5] provided a complete classification on the global dynamics of system (1.1), which says that either one of the two semi-trivial steady states is globally asymptotically stable, or there is a unique co-existence steady state which is globally asymptotically stable, or the system is degenerate in the sense that there is a compact global attractor consisting of a continuum of steady states which connect the two semi-trivial steady states. This result, in particular, completely settles the conjecture proposed by Lou in [11], and also, encompasses the widely accepted result "slower diffuser always wins" [4]; indeed, it put these two results in perspective. We refer the interested readers to [6, 7] for some further developments. See also [10, 12] for some previous discussion on this topic.

For system (1.2), indeed it is far from being completely understood although a recent progress was made by Tang and Wang [17]. The main results in [17] can be summarized as follows. Assuming that

$$r(x)\in C^2(\overline{\Omega}) \text{ and } r(x)\ge 0 \text{ is not constant in } \Omega, \tag{H_0}$$

and

$$d_i(r)\in C^2([0,+\infty)]), d_i(r)>0 \text{ and } d_i'(r)\le 0 \text{ on } [0,+\infty). \tag{H_1}$$

Theorem 1.1 ([17]) *Let $d_1(r)$ and $d_2(r)$ satisfy (H_1) with r fulfilling (H_0). If $d_1(r)=\vartheta d_2(r)$ for some constant $\vartheta>0$ and $d_2(r)r$ is not constant in Ω, then the semi-trivial steady state $(\theta_1,0)$ (resp. $(0,\theta_2)$) of system (1.2) is globally asymptotically stable for any $\vartheta\mu_1<\mu_2$ (resp. $\vartheta\mu_1>\mu_2$).*

Theorem 1.2 ([17]) *If $d_i(r)(i=1,2)$ satisfies (H_1) with r fulfilling (H_0), and $d_1(r)\not\equiv Cd_2(r)$ for any constant $C>0$, then the following results hold.*

(i) If $\delta_1<0<\delta_2$ (resp. $\delta_2<0<\delta_1$), then the system (1.2) has a globally asymptotically stable steady state $(\theta_1,0)$ (resp. $(0,\theta_2)$) for large μ_1,μ_2.

(ii) If $\delta_1,\delta_2>0$, then the system (1.2) admits a globally asymptotically stable coexistence steady state (U_*,V_*) for large μ_1,μ_2. Furthermore, the coexistence steady state (U_*,V_*) converges to $(\frac{\delta_1}{c_0},\frac{\delta_2}{c_0})$ in $C^1(\Omega)$ as $\mu_1,\mu_2\to+\infty$, where δ_1,δ_2,c_0 are defined as follows:

$$\begin{cases}\delta_1:=\int_\Omega rd_2(r)^{-1}\mathrm{d}x\int_\Omega d_1(r)^{-2}\mathrm{d}x-\int_\Omega rd_1(r)^{-1}\mathrm{d}x\int_\Omega (d_1(r)d_2(r))^{-1}\mathrm{d}x,\\ \delta_2:=\int_\Omega rd_1(r)^{-1}\mathrm{d}x\int_\Omega d_2(r)^{-2}\mathrm{d}x-\int_\Omega rd_2(r)^{-1}\mathrm{d}x\int_\Omega (d_1(r)d_2(r))^{-1}\mathrm{d}x,\\ c_0=\int_\Omega d_1(r)^{-2}\mathrm{d}x\int_\Omega d_2(r)^{-2}\mathrm{d}x-(\int_\Omega (d_1(r)d_2(r))^{-1}\mathrm{d}x)^2.\end{cases} \tag{1.3}$$

The main purpose of this paper is to obtain a more comprehensive understanding on the dynamics of system (P) with more general diffusion coefficients μ_1,μ_2.

1.2 Main Results

Coming back to system (P), we note that, owing to the resource-dependent dispersal rate functions $d_i(r_i)(i = 1, 2)$, it is inconvenient to consider (P) directly. Therefore, to investigate the dynamics of system (P), throughout this paper, we borrow the following changes of the variables(see [17]):

$$u = d_1(r_1)u_1, v = d_2(r_2)u_2, \tag{1.4}$$

and make the following basic hypotheses:

$$r_1(x), r_2(x) > 0, d_1(r_1), d_2(r_2) > 0. \tag{1.5}$$

Based on the above changes, system (P) is reduced to

$$\begin{cases} u_t = \mu_1 d_1(r_1)\Delta u + u(r_1(x) - \dfrac{u}{d_1(r_1)} - \dfrac{v}{d_2(r_2)}), & \text{in } \Omega \times (0, T), \\ v_t = \mu_2 d_2(r_2)\Delta v + v(r_2(x) - \dfrac{u}{d_1(r_1)} - \dfrac{v}{d_2(r_2)}), & \text{in } \Omega \times (0, T), \\ \dfrac{\partial u}{\partial n} = \dfrac{\partial v}{\partial n} = 0, & \text{on } \partial\Omega \times (0, T), \\ u(x, 0) = u_0(x) \geq, \not\equiv 0, & \text{in } \Omega, \\ v(x, 0) = v_0(x) \geq, \not\equiv 0, & \text{in } \Omega. \end{cases} \tag{1.6}$$

Before describing precisely the outcome of our paper, we first introduce some notations:

$$\begin{aligned} &\mathcal{Q} := (0, +\infty) \times (0, +\infty); \\ &\Sigma_u := \{(\mu_1, \mu_2) \in \mathcal{Q} : (\theta_1, 0) \text{ is linearly stable}\}; \\ &\Sigma_v := \{(\mu_1, \mu_2) \in \mathcal{Q} : (0, \theta_2) \text{ is linearly stable}\}; \\ &\Sigma_0 := \{(\mu_1, \mu_2) \in \mathcal{Q} : \text{ both } (\theta_1, 0) \text{ and} (0, \theta_2) \text{ are linearly unstable}\}. \end{aligned} \tag{1.7}$$

It is known that system (1.6) generates a monotone dynamical system and the following statements concerning the global dynamics of system (1.6) hold.

Proposition 1.1 *With the hypotheses* (1.5), *the following results hold.*

(*i*) *If a steady state of system* (1.6) *is linearly stable* (*resp. linearly unstable*), *then it is locally asymptotically stable* (*resp. unstable*).

(*ii*) *If system* (1.6) *admits two semi-trivial steady states* $(\theta_1, 0)$ *and* $(0, \theta_2)$, *and does not admit any coexistence steady state*, *then one of the semi-trivial steady states is globally asymptotically stable and the other one is unstable.*

(*iii*) *If system* (1.6) *admits two linearly unstable*(*resp. stable*) *semi-trivial steady states*, *then it admits at least one locally asymptotically stable*(*resp. unstable*) *coexistence steady state.*

(*iv*) *If every coexistence steady state of system* (1.6) *is linearly stable*, *then either there are no coexistence steady states and one of the two semi-trivial steady state is globally asymptotically stable while the other one is unstable*, *or there is a unique coexistence steady state which is globally asymptotically stable.*

We are now in a position to state our main results. Note that the local (in time) existence and uniqueness of nonnegative classical solution of (1.6) are rather standard. We will concentrate here on the long time behaviors of such solutions.

Throughout our paper, we use d_i to denote $d_i(r_i)$ for simplicity.

Theorem 1.3 *Suppose* (1.5) *holds, and* $r_1(x) - r_2(x)$ *changes sign. Then for all* μ_1 *and* μ_2 *sufficiently small, both the semi-trivial steady states* $(\theta_1, 0)$ *and* $(0, \theta_2)$ *of system* (1.6) *are linearly unstable and there is at least one stable coexistence steady state.*

Next theorem is about the dynamics of (1.6) for large μ_1 and/or μ_2. To state next theorem precisely, we introduce the following notations.

$$\begin{cases} a_1 := \int_\Omega \frac{r_1}{d_1}\mathrm{d}x \int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x,\ a_2 := \int_\Omega \frac{r_2}{d_2}\mathrm{d}x \int_\Omega \frac{1}{d_1^2}\mathrm{d}x,\ a_3 := \int_\Omega \frac{r_1}{d_1}\mathrm{d}x \int_\Omega \frac{1}{d_2^2}\mathrm{d}x, \\ a_4 := \int_\Omega \frac{r_2}{d_2}\mathrm{d}x \int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x, \\ a_5 := \frac{\int_\Omega r_1 d_1^{-1}\mathrm{d}x}{\int_\Omega d_1^{-2}\mathrm{d}x},\ a_6 := \frac{\int_\Omega r_2 d_1 \mathrm{d}x}{|\Omega|},\ a_7 := \frac{\int_\Omega r_2 d_2^{-1}\mathrm{d}x}{\int_\Omega d_2^{-2}\mathrm{d}x},\ a_8 := \frac{\int_\Omega r_1 d_2 \mathrm{d}x}{|\Omega|}. \end{cases} \tag{1.8}$$

For $i = 1, 2$, denote by ρ_i, γ_i the unique solution satisfying

$$\begin{cases} \Delta\rho_i + c_{r_i}\left(\frac{r_i}{d(r_i)} - \frac{c_{r_i}}{d^2(r_i)}\right) = 0,\ x \in \Omega, \\ \frac{\partial \rho_i}{\partial n} = 0,\ x \in \partial\Omega, \\ \int_\Omega \rho_i \mathrm{d}x = 0, \end{cases} \tag{1.9}$$

$$\begin{cases} \Delta\gamma_i + \left(\frac{r_i}{d(r_i)} - \frac{2c_{r_i}}{d^2(r_i)}\right)(\rho_i + C_i) = 0,\ x \in \Omega, \\ \frac{\partial \gamma_i}{\partial n} = 0,\ x \in \partial\Omega, \\ \int_\Omega \gamma_i \mathrm{d}x = 0, \\ C_i = \frac{\int_\Omega \rho_i \left(\frac{r_i}{d(r_i)} - \frac{2c_{r_i}}{d^2(r_i)}\right)\mathrm{d}x}{c_{r_i}\int_\Omega \frac{1}{d^2(r_i)}\mathrm{d}x}, \end{cases} \tag{1.10}$$

where c_{r_i} are defined in (2.3).

Theorem 1.4 *Suppose* (1.5) *holds. Then the following hold for system* (1.6).

(*i*) *If* $a_1 < a_2$ *and* $a_3 > a_4$, *then both the semi-trivial steady state* $(\theta_1, 0)$ *and* $(0, \theta_2)$ *of system* (1.6) *are linearly unstable and* (1.6) *has a coexistence steady state which is globally asymptotically stable for* μ_1, μ_2 *sufficiently large.*

(*ii*) *If* $a_1 > a_2$, $a_3 > a_4$, $a_5 < a_6$, $a_7 < a_8$, *then for* μ_1, μ_2 *sufficiently large,* $(\theta_1, 0)$ *is globally asymptotically stable.*

(*iii*) *If* $a_1 < a_2$, $a_3 < a_4$, $a_5 < a_6$, $a_7 < a_8$, *then for* μ_1, μ_2 *sufficiently large,* $(0, \theta_2)$ *is globally asymptotically stable.*

(*iv*) *If* $a_1 = a_2$, $a_3 > a_4$, $a_5 < a_6$, $a_7 < a_8$, $\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x < 0$, *then* $(\theta_1, 0)$ *is linearly unstable for all sufficiently large* μ_1 *and* $\mu_2 > 0$; $(0, \theta_2)$ *is linearly unstable for all sufficiently large* μ_2 *and* $\mu_1 > 0$. *Furthermore, there exists at least one stable coexistence steady state for all sufficiently large* μ_1, μ_2.

(v) *If* $a_1 = a_2$, $a_3 > a_4$, $a_5 < a_6$, $a_7 < a_8$, $\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2} dx > 0$, *then there exists continuous function* $\tilde{\mu}_2^*(\mu_1)$ *defined for all sufficiently large* μ_1 *such that*

$$\begin{cases} \Sigma_u \cap \{(\mu_1, \mu_2) \in \mathcal{Q} : \mu_1 \text{ is large enough}\} \\ \quad = \{(\mu_1, \mu_2) : \mu_1 \text{ is large enough}, \mu_2 > \tilde{\mu}_2^*(\mu_1)\}, \\ \tilde{\mu}_2^*(\mu_1) = \frac{\int_\Omega \frac{\rho_2}{c_{r_2}}(\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2})}{\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2} dx} \mu_1 + B_1 + O(\frac{1}{\mu_1}) \text{ for all sufficiently large } \mu_1, \end{cases} \tag{1.11}$$

where B_1 *is a constant. Moreover* $(0, \theta_2)$ *is linearly unstable for all* $\mu_2 > \mu$ *and* $\mu_1 > 0$.

(vi) *If* $a_1 < a_2$, $a_3 = a_4$, $a_5 < a_6$, $a_7 < a_8$, $\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2} dx < 0$, *then* $(\theta_1, 0)$ *is linearly unstable for all sufficiently large* μ_1 *and* $\mu_2 > 0$; $(0, \theta_2)$ *is linearly unstable for all sufficiently large* μ_2 *and* $\mu_1 > 0$. *Furthermore, there exists at least one stable coexistence steady state for all sufficiently large* μ_1, μ_2.

(vii) *If* $a_1 < a_2$, $a_3 = a_4$, $a_5 < a_6$, $a_7 < a_8$, $\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2} dx > 0$, *then there exists continuous function* $\tilde{\mu}_1^*(\mu_2)$ *defined for all sufficiently large* μ_2 *such that*

$$\begin{cases} \Sigma_v \cap \{(\mu_1, \mu_2) \in \mathcal{Q} : \mu_2 \text{ is large enough}\} \\ \quad = \{(\mu_1, \mu_2) : \mu_2 \text{ is large enough}, \mu_1 > \tilde{\mu}_1^*(\mu_2)\}, \\ \tilde{\mu}_1^*(\mu_2) = \frac{\int_\Omega \frac{\rho_1}{c_{r_1}}(\frac{r_1}{d_1} - \frac{c_{r_1}}{d_1 d_2})}{\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2} dx} \mu_2 + B_2 + O(\frac{1}{\mu_2}) \text{ for all sufficiently large } \mu_2, \end{cases} \tag{1.12}$$

where B_2 *is a constant. Moreover* $(\theta_1, 0)$ *is linearly unstable for all sufficiently large* μ_1 *and* $\mu_2 > 0$.

This paper is organized as follows. In Sect. 2, we provide some preliminary results. In Sect. 3, we prove Theorem 1.3 and Theorem 1.4.

2 Preliminaries

In this section, we primarily give some useful results which will be utilized in later analysis.

2.1 Single Equation Problem

Let us consider the following elliptic boundary value problem.

$$\begin{cases} \mu d(r)\Delta\theta + \theta(r(x) - \frac{\theta}{d(r)}) = 0, \ x \in \Omega, \\ \frac{\partial \theta}{\partial n} = 0, \ x \in \partial\Omega, \end{cases} \tag{2.1}$$

where $r(x)$ satisfies (H_0) and $d(r)$ satisfies (H_1). Under these conditions, it follows from [17, Lemma 2.1] that problem (2.1) always has a positive solution, which implies the existence of semi-trivial steady state $(\theta_1, 0)$ and $(0, \theta_2)$ for system (1.6).

The following statements can be followed by [17].

Lemma 2.1 ([17]) *If $d(r)$ satisfies (H_1) with r fulfilling (H_0), then the unique positive solution θ of problem (2.1) satisfies*

$$\begin{cases} \min\limits_{x\in\bar{\Omega}}(d(r)r) \le \theta \le \max\limits_{x\in\bar{\Omega}}(d(r)r), \\ \lim\limits_{\mu\to 0} \|\theta - rd(r)\|_\infty = 0, \\ \lim\limits_{\mu\to+\infty} \|\theta - c_r\|_{C^1(\Omega)} = 0, \end{cases} \tag{2.2}$$

where

$$c_r = \frac{\int_\Omega r d(r)^{-1} \mathrm{d}x}{\int_\Omega d(r)^{-2} \mathrm{d}x}. \tag{2.3}$$

2.2 Eigenvalue Problem

To study the stability of steady states, we introduce the following linear eigenvalue problem, which will play a very important role.

$$\begin{cases} \mu D(x)\Delta\phi + m(x)\phi = \sigma\phi, \ x \in \Omega, \\ \dfrac{\partial\phi}{\partial n} = 0, \ x \in \partial\Omega, \end{cases} \tag{2.4}$$

where $D(x) \in C(\bar{\Omega}; (0, +\infty))$, $m(x) \in C(\bar{\Omega}; \mathbb{R})$. By the Krein-Rutman Theorem [9], problem (2.4) admits a principal eigenvalue, denoted by $\sigma_1(\mu D, m)$ with a strictly positive eigenfunction $\phi_1(\mu D, m)$ in $\bar{\Omega}$. Moreover, by the variational approach, $\sigma_1(\mu D, m)$ can be characterized by

$$\sigma_1(\mu D, m) = \sup_{0\neq\phi\in H^1(\Omega)} \frac{\int_\Omega (-\mu|\nabla\phi|^2 + \frac{m(x)}{D(x)}\phi^2)\mathrm{d}x}{\int_\Omega \frac{\phi^2}{D(x)}\mathrm{d}x}. \tag{2.5}$$

Then some properties of $\sigma_1(\mu D, m)$ hold as follows.

Proposition 2.1 ([17]) *The following statements on $\sigma_1(\mu D, m)$ are true.*

(1) *$\sigma_1(\mu D, m)$ and $\phi_1(\mu D, m)$ depend continuously and differentially on parameters μ;*

(2) *$\sigma_1(\mu D, m)$ is strictly decreasing in the parameter μ provided m is non-constant;*

(3) *$\sigma_1(\mu D, m)$ is strictly increasing in the weight function $m(x)$, that is, if $m_1(x) \le, \not\equiv m_2(x)$ in $\bar{\Omega}$, then $\sigma_1(\mu D, m_1) < \sigma_1(\mu D, m_2)$;*

(4) $\lim\limits_{\mu\to 0} \sigma_1(\mu D, m) = \max\limits_{x\in\bar{\Omega}} m(x)$ *and* $\lim\limits_{\mu\to+\infty} \sigma_1(\mu D, m) = \dfrac{\int_\Omega \frac{m}{D(x)}\mathrm{d}x}{\int_\Omega \frac{1}{D(x)}\mathrm{d}x}$.

Indeed, by the same spirit as in [10, Corollary 2.10], one can get that the linear stability of $(\theta_1, 0)$ and $(0, \theta_2)$ can be determined, respectively, by the sign of $\sigma_1(\mu_2 d_2, r_2 - \dfrac{\theta_1}{d_1})$ and $\sigma_1(\mu_1 d_1, r_1 - \dfrac{\theta_2}{d_2})$. More precisely, $(\theta_1, 0)$ is linearly stable(resp. linearly unstable) provided $\sigma_1(\mu_2 d_2, r_2 - \dfrac{\theta_1}{d_1}) < 0$ (resp. $\sigma_1(\mu_2 d_2, r_2 - \dfrac{\theta_1}{d_1}) > 0$). The stability of $(0, \theta_2)$ can be described in a similar manner. Furthermore, if a steady state is linearly stable (resp. linearly unstable), then it should be asymptotically stable (resp. unstable) (see, e.g., [16, Theorem 7.6.2]).

In order to further characterize properties of $\sigma_1(\mu D, m)$, the following eigenvalue problem with indefinite weight is needed:

$$\begin{cases} D(x)\Delta\varphi + \lambda m(x)\varphi = 0, \ x \in \Omega, \\ \dfrac{\partial\varphi}{\partial n} = 0, \ x \in \partial\Omega, \end{cases} \tag{2.6}$$

where $D(x) \in L^\infty(\bar{\Omega}; (0, +\infty))$, $m(x) \in L^\infty(\Omega)$ is non-constant and could change sign in Ω. We say that λ is a principal eigenvalue if (2.6) has a positive solution. (Notice that 0 is always a principal eigenvalue.) Now we can state the following result, which can be deduced by the same arguments as in [1, 8, 13].

Proposition 2.2 *The problem (2.6) has a nonzero principal eigenvalue $\lambda_1 = \lambda_1(D, m)$ if and only if m changes sign and $\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x \neq 0$. More precisely, if m changes sign, then*

(i) $\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x = 0 \Leftrightarrow 0$ *is the only principal eigenvalue*;
(ii) $\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x > 0 \Leftrightarrow \lambda_1(D, m) < 0$;
(iii) $\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x < 0 \Leftrightarrow \lambda_1(D, m) > 0$;
(iv) $\lambda_1(D, m_1) > \lambda_1(D, m_2)$ *if* $m_1 \leq m_2$, $m_1 \not\equiv m_2$ *a.e., and* m_1, m_2 *both change sign*;
(v) $\lambda_1(D, m)$ *is continuous in* m.

We now state the following proposition which collects some important relationships between $\lambda_1(D, m)$ and $\sigma_1(\mu D, m)$. For a proof, see, e.g., [2, 14].

Proposition 2.3 *The first eigenvalue $\sigma_1(\mu D, m)$ of (2.4) depends smoothly on $\mu > 0$ and continuously on $m \in L^\infty(\Omega)$. Moreover, it has the following properties*:

(i) $\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x \geq 0$ *and* $m \not\equiv 0 \Rightarrow \sigma_1(\mu D, m) > 0$ *for all* $\mu > 0$;
(ii)

$$\int_\Omega \frac{m(x)}{D(x)}\mathrm{d}x < 0 \text{ and } m \text{ changes sign in } \Omega \Rightarrow \begin{cases} \sigma_1(\mu D, m) > 0 \text{ for all } \mu < \dfrac{1}{\lambda_1(D, m)}, \\ \sigma_1(\mu D, m) = 0 \text{ if } \mu = \dfrac{1}{\lambda_1(D, m)}, \\ \sigma_1(\mu D, m) < 0 \text{ for all } \mu > \dfrac{1}{\lambda_1(D, m)}. \end{cases}$$

3 Proof of Main Results

This section is devoted to the establishment of our main conclusions.

3.1 Proof of Theorem 1.3

Here, we shall show a result in the case when both μ_1 and μ_2 are small.

Proof of Theorem 1.3 By Proposition 2.2, one can define

$$\tilde{\mu}_2^*(\mu_1) = \begin{cases} +\infty, \text{ if } \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x \geq 0, \\ \dfrac{1}{\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1})}, \text{ if } \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x < 0. \end{cases} \tag{3.1}$$

Note from Proposition 2.3 that $\sigma_1(\mu_2 d_2, r_2 - \frac{\theta_1}{d_1}) > 0$ for $0 < \mu_2 < \tilde{\mu}_2^*(\mu_1)$. By Lemma 2.1, we have that $\lim\limits_{\mu_1 \to 0} \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x = \int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x$. This implies that if $\int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x > 0$, then $\int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x > 0$ and $\tilde{\mu}_2^*(\mu_1) = +\infty$ for μ_1 small; if $\int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x < 0$, then $\lim\limits_{\mu_1 \to 0} \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x = \int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x < 0$. For the case $\int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x = 0$, let $\{\mu_{1,k}\}_{k=1}^\infty$ be a sequence such that $\mu_{1,k} \to 0$ as $k \to +\infty$. Since $\int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x\big|_{\mu_1 = \mu_{1,k}} \geq 0$ leads to $\tilde{\mu}_2^*(\mu_1) = +\infty$, without loss of generality, we may assume that $\int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x\big|_{\mu_1 = \mu_{1,k}} < 0$ for all $k \geq 1$. Then by Proposition 2.2(i) and similar arguments as in the proof of [13, Lemma 2.4], we can get that $\lim\limits_{\mu_1 \to 0} \lambda_1(d_2, r_2 - \frac{\theta_1}{d_1}) = 0$. Therefore we arrive at

$$\lim_{\mu_1 \to 0} \tilde{\mu}_2^*(\mu_1) = \begin{cases} +\infty, \text{ if } \int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x \geq 0, \\ \dfrac{1}{\lambda_1(d_2, r_2 - r_1)} > 0, \text{ if } \int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x < 0. \end{cases} \tag{3.2}$$

It then follows from Proposition 2.3(i) that if $\int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x \geq 0$, $\lim\limits_{\mu_1 \to 0} \sigma_1(\mu_2 d_2, r_2 - \frac{\theta_1}{d_1}) = \sigma_1(\mu_2 d_2, r_2 - r_1) > 0$ for any $d_2 > 0$ since $r_1(x) \not\equiv r_2(x)$. While if $\int_\Omega (\frac{r_2}{d_2} - \frac{r_1}{d_2})\mathrm{d}x < 0$, then $\mu_2 < \dfrac{1}{\lambda_1(d_2, r_2 - r_1)}$ for sufficiently small μ_1. Therefore, Proposition 2.3(ii) yields that $\lim\limits_{\mu_1 \to 0} \sigma_1(\mu_2 d_2, r_2 - \frac{\theta_1}{d_1}) = \sigma_1(\mu_2 d_2, r_2 - r_1) > 0$ since $\mu_2 < \dfrac{1}{\lambda_1(d_2, r_2 - r_1)}$. Hence, one has that $\sigma_1(\mu_2 d_2, r_2 - \frac{\theta_1}{d_1}) > 0$ provided μ_1 and μ_2 are small enough. Similarly, we can get that $\sigma_1(\mu_1 d_1, r_1 - \frac{\theta_2}{d_2}) > 0$ for μ_1 and μ_2 sufficiently small. We then conclude that $(\theta_1, 0)$, $(0, \theta_2)$ are unstable and then system (1.6) has a stable coexistence steady state for all μ_1 and μ_2 sufficiently small. □

3.2 Proof of Theorem 1.4

Before verify Theorem 1.4, we first give a asymptotic expansion of the unique solution θ of (2.1) for μ large. Denote by ρ the unique solution of

$$\begin{cases} \Delta\rho + c_r(\dfrac{r}{d(r)} - \dfrac{c_r}{d^2(r)}) = 0, \ x \in \Omega, \\ \dfrac{\partial\rho}{\partial n} = 0, \ x \in \partial\Omega, \\ \displaystyle\int_\Omega \rho \mathrm{d}x = 0, \end{cases} \tag{3.3}$$

and let γ be the unique solution of

$$\begin{cases} \Delta\gamma + (\dfrac{r}{d(r)} - \dfrac{2c_r}{d^2(r)})(\rho + C) = 0, \ x \in \Omega, \\ \dfrac{\partial\gamma}{\partial n} = 0, \ x \in \partial\Omega, \\ \displaystyle\int_\Omega \gamma \mathrm{d}x = 0, \\ C = \dfrac{\int_\Omega \rho(\frac{r}{d(r)} - \frac{2c_r}{d^2(r)})\mathrm{d}x}{c_r \int_\Omega \frac{1}{d^2(r)}\mathrm{d}x}, \end{cases} \tag{3.4}$$

where c_r is defined in Lemma 2.1.

Proposition 3.1 *There exists a constant* $\mu > 0$ *such that*

$$\theta = c_r + \frac{\rho + C}{\mu} + \frac{\gamma + E}{\mu^2} + O(\frac{1}{\mu^3}) \text{ for all } \mu > \mu, \tag{3.5}$$

where ρ, C *and* γ *are defined in* (3.3) *and* (3.4) *respectively,*

$$E := \frac{\int_\Omega \rho^2(\frac{3}{d^2(r)} - \frac{r}{dc_r})\mathrm{d}x - 2C\int_\Omega \frac{\rho}{d^2(r)}\mathrm{d}x - \int_\Omega \frac{c_r\gamma}{d^2(r)}\mathrm{d}x}{c_r \int_\Omega \frac{1}{d^2(r)}\mathrm{d}x}. \tag{3.6}$$

Proof We shall apply the similar ideas in [6] to show this proposition. Note that the definition of C guarantees the existence and uniqueness of the solution to (3.4). Multiplying the equation of ρ (resp. γ) by γ (resp. ρ) respectively and integrating over Ω by parts, there holds

$$c_r \int_\Omega (\frac{r}{d(r)} - \frac{c_r}{d^2(r)})\gamma \mathrm{d}x = \int_\Omega (\frac{r}{d(r)} - \frac{2c_r}{d^2(r)})\rho^2 \mathrm{d}x + C^2 c_r \int_\Omega \frac{1}{d^2(r)}\mathrm{d}x. \tag{3.7}$$

By some calculations, we hence get that

$$\int_\Omega (\frac{r}{d(r)} - \frac{2c_r}{d^2(r)})(\gamma + E)\mathrm{d}x - \int_\Omega \frac{(\rho + C)^2}{d^2(r)}\mathrm{d}x = 0, \tag{3.8}$$

which guarantees the existence and uniqueness of the solution $\theta^{(3)}$ to the problem

$$\begin{cases} \Delta\theta^{(3)} + (\dfrac{r}{d(r)} - \dfrac{2c_r}{d^2(r)})(\gamma + E) - \dfrac{(\rho + C)^2}{d^2(r)} = 0, \ x \in \Omega, \\ \dfrac{\partial\theta^{(3)}}{\partial n} = 0, \ x \in \partial\Omega, \\ \displaystyle\int_\Omega \theta^{(3)} \mathrm{d}x = 0. \end{cases} \tag{3.9}$$

Let $\theta^{(4)}$ satisfies

$$\begin{cases} \Delta\theta^{(4)} + (\dfrac{r}{d(r)} - \dfrac{2c_r}{d^2(r)})(\theta^{(3)} + F) - \dfrac{2(\rho + C)(\gamma + E)}{d^2(r)} = 0, \ x \in \Omega, \\ \dfrac{\partial\theta^{(4)}}{\partial n} = 0, \ x \in \partial\Omega, \\ \displaystyle\int_\Omega \theta^{(4)} \mathrm{d}x = 0, \end{cases} \tag{3.10}$$

where F is the unique number such that $\theta^{(4)}$ exists. It then indicates that for all μ sufficiently large

$$\theta^{\pm} := c_r + \frac{\rho + C}{\mu} + \frac{\gamma + E}{\mu^2} + \frac{\theta^{(3)} + F}{\mu^3} + \frac{\theta^{(4)}}{\mu^4} \pm (\frac{1}{\mu^3} + \frac{\rho}{c_r\mu^4}) \tag{3.11}$$

satisfies

$$\begin{cases} 0 < \theta^- < \theta^+, \\ \mu d(r)\Delta\theta^{\pm} + \theta^{\pm}(r - \dfrac{\theta^{\pm}}{d(r)}) = -(\pm\dfrac{c_r}{d^2(r)\mu^3}) + O(\dfrac{1}{\mu^4}) \end{cases} \tag{3.12}$$

by some calculations. Hence $\theta^{\pm}$ is a pair of upper and lower solutions to (2.1) for large μ. Therefore $\theta^- < \theta < \theta^+$ by the upper/lower solution method and the uniqueness of θ. The proof of this proposition is hence completed. □

Now we begin to prove Theorem 1.4

Proof of Theorem 1.4 We will divide the proof into several cases.

Case 1. $a_1 < a_2$ and $a_3 > a_4$.

Notice that in this case, $d_1(r_1) \not\equiv Cd_2(r_2)$ for any constant $C > 0$. From Lemma 2.1 one can have that

$$\begin{aligned} &\lim_{\mu_2 \to +\infty} \int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x = \int_\Omega (\frac{r_1}{d_1} - \frac{c_{r_2}}{d_1 d_2})\mathrm{d}x = \int_\Omega \frac{r_1}{d_1}\mathrm{d}x - c_{r_2}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \\ &= \int_\Omega \frac{r_1}{d_1}\mathrm{d}x - \frac{\int_\Omega r_2 d_2^{-1}\mathrm{d}x}{\int_\Omega d_2^{-2}\mathrm{d}x}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x = \frac{a_3 - a_4}{\int_\Omega d_2^{-2}\mathrm{d}x} > 0. \end{aligned} \tag{3.13}$$

This shows that $\displaystyle\int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x > 0$ for μ_2 sufficiently large. It then follows from Proposition 2.3(i) that $\sigma_1(\mu_1 d_1, r_1 - \dfrac{\theta_2}{d_2}) > 0$ for μ_2 sufficiently large. Combined with $a_1 < a_2$, $\sigma_1(\mu_2 d_2, r_2 - \dfrac{\theta_1}{d_1}) > 0$ for large μ_1 can be handled similarly. Hence, both $(\theta_1, 0)$ and $(0, \theta_2)$

are linearly unstable and moreover by Proposition 1.1(iii), (1.6) has at least a locally asymptotically stable coexistence steady state (u^*, v^*) for μ_1, μ_2 sufficiently large.

Next, we claim that, for any coexistence steady state (u^*, v^*), we have, as $\mu_1, \mu_2 \to +\infty$,

$$(u^*, v^*) \to \left(\frac{a_3 - a_4}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x \int_\Omega \frac{1}{d_2^2}\mathrm{d}x - (\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x)^2}, \frac{a_2 - a_1}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x \int_\Omega \frac{1}{d_2^2}\mathrm{d}x - (\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x)^2}\right). \tag{3.14}$$

It follows readily from the maximum principle that both u^*, v^* are uniformly bounded, independent of μ_1, μ_2. Passing to a subsequence if necessary, we have $(u^*, v^*) \to (\tilde{u}, \tilde{v})$ in $C^2(\bar{\Omega})$ as $\mu_1, \mu_2 \to +\infty$, where $\tilde{u}$ and $\tilde{v}$ are two nonnegative constants, by elliptic regularity estimates. Dividing the equations for u^*, v^* by $d_1(r_1)$, $d_2(r_2)$ respectively, and then integrating the resulting equations over Ω, we obtain after passing to the limit that

$$\int_\Omega \tilde{u}\left(\frac{r_1}{d_1} - \frac{\tilde{u}}{d_1^2} - \frac{\tilde{v}}{d_1 d_2}\right)\mathrm{d}x = 0 = \int_\Omega \tilde{v}\left(\frac{r_2}{d_2} - \frac{\tilde{u}}{d_1 d_2} - \frac{\tilde{v}}{d_2^2}\right)\mathrm{d}x, \tag{3.15}$$

i.e.,

$$\begin{cases} \tilde{u}\left(\int_\Omega \frac{r_1}{d_1}\mathrm{d}x - \tilde{u}\int_\Omega \frac{1}{d_1^2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x\right) = 0, \\ \tilde{v}\left(\int_\Omega \frac{r_2}{d_2}\mathrm{d}x - \tilde{u}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{1}{d_2^2}\mathrm{d}x\right) = 0. \end{cases} \tag{3.16}$$

Case ①. $\tilde{u} = 0 = \tilde{v}$.

This implies that u^* and $v^* \to 0$ in $C^2(\bar{\Omega})$, and consequently, for μ_1, μ_2 sufficiently large,

$$\int_\Omega u^*\left(\frac{r_1}{d_1} - \frac{u^*}{d_1^2} - \frac{v^*}{d_1 d_2}\right)\mathrm{d}x > 0, \tag{3.17}$$

a contradiction.

Case ②. $\tilde{u} = 0$, $\tilde{v} > 0$.

From the second equation in (3.16) it follows that $\tilde{v} = a_7$. Setting $w = \dfrac{u^*}{\|u^*\|_\infty}$, we have $\|w\|_\infty = 1$ and

$$\begin{cases} \Delta w = -\dfrac{w}{\mu_1 d_1}\left(r_1(x) - \dfrac{u^*}{d_1} - \dfrac{v^*}{d_2}\right) \text{ in } \Omega, \\ \dfrac{\partial w}{\partial n} = 0 \text{ on } \partial\Omega. \end{cases} \tag{3.18}$$

As $\mu_1 \to +\infty$, from elliptic regularity estimates we conclude that $w \to 1$ on $\bar{\Omega}$. Now

$$0 = \int_\Omega \frac{w}{d_1}\left(r_1(x) - \frac{u^*}{d_1} - \frac{v^*}{d_2}\right)\mathrm{d}x \to \int_\Omega \frac{1}{d_1}\left(r_1(x) - \frac{\tilde{v}}{d_2}\right)\mathrm{d}x. \tag{3.19}$$

Therefore, $\tilde{v} = \dfrac{\int_\Omega r_1 d_1^{-1}\mathrm{d}x}{\int_\Omega d_1^{-1} d_2^{-1}\mathrm{d}x}$, a contradiction.

Case ③. $\tilde{u} > 0$, $\tilde{v} = 0$. This case may be handled in a similar fashion as Case ②.

Case ④. $\tilde{u} > 0$, $\tilde{v} > 0$. Now it follows from (3.16) that

$$\tilde{u} = \frac{a_3 - a_4}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x \int_\Omega \frac{1}{d_2^2}\mathrm{d}x - (\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x)^2}, \tilde{v} = \frac{a_2 - a_1}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x \int_\Omega \frac{1}{d_2^2}\mathrm{d}x - (\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x)^2}, \tag{3.20}$$

and our assertion (3.14) is established.

It remains to show that every coexistence steady state (u^*, v^*) is stable if both μ_1 and μ_2 are large. Consider the linearized eigenvalue problem at (u^*, v^*):

$$\begin{cases} \mu_1 d_1(r_1)\Delta\phi + \phi\big(r_1(x) - \dfrac{2u^*}{d_1(r_1)} - \dfrac{v^*}{d_2(r_2)}\big) - \dfrac{u^*}{d_2(r_2)}\psi + \lambda\phi = 0, \text{ in } \Omega, \\ \mu_2 d_2(r_2)\Delta\psi + \psi\big(r_2(x) - \dfrac{u^*}{d_1(r_1)} - \dfrac{2v^*}{d_2(r_2)}\big) - \dfrac{v^*}{d_1(r_1)}\phi + \lambda\psi = 0, \text{ in } \Omega, \\ \dfrac{\partial\phi}{\partial n} = 0 \text{ on } \partial\Omega, \\ \dfrac{\partial\psi}{\partial n} = 0 \text{ on } \partial\Omega. \end{cases} \tag{3.21}$$

By the Krein–Rutman theory, (3.21) has a real principal eigenvalue λ_1, and all other eigenvalues λ have real parts $\geq \lambda_1$. Assume for contradiction that the principal eigenvalue $\lambda_1 \leq 0$. Denote the corresponding eigenfunction by (ϕ, ψ), normalized by $\|\dfrac{\phi^2}{d_1}\|_1 + \|\dfrac{\psi^2}{d_2}\|_1 = 1$. Then,

$$\begin{aligned} \lambda_1 &= \mu_1\|\nabla\phi\|_2^2 - \int_\Omega \phi^2\big(\frac{r_1}{d_1} - \int_\Omega \frac{2u^*}{d_1^2} - \frac{v^*}{d_1 d_2}\big)\mathrm{d}x + \int_\Omega \frac{u^*}{d_1 d_2}\phi\psi\,\mathrm{d}x \\ &+\mu_2\|\nabla\psi\|_2^2 - \int_\Omega \psi^2\big(\frac{r_2}{d_2} - \frac{u^*}{d_1 d_2} - \frac{2v^*}{d_2^2}\big)\mathrm{d}x + \int_\Omega \frac{v^*}{d_1 d_2}\phi\psi\,\mathrm{d}x \geq -M, \end{aligned} \tag{3.22}$$

where M is a constant independent of μ_1, μ_2; i.e., $0 \geq \lambda_1 \geq -M$.

Now, letting $\mu_1, \mu_2 \to +\infty$ in (3.21), by elliptic regularity estimates, we have, by passing to a subsequence if necessary, $\lambda_1 \to \tilde{\lambda}_1 \leq 0$, $(\phi, \psi) \to (\tilde{\phi}, \tilde{\psi})$, where $\tilde{\phi}$ and $\tilde{\psi}$ are two constants. Now, dividing the two equations in (3.21) by $d_1(r_1)$, $d_2(r_2)$ respectively, integrating the resulting equations and then passing to limits, we obtain

$$\begin{cases} \tilde{\phi}\big(\displaystyle\int_\Omega \frac{r_1}{d_1}\mathrm{d}x - \tilde{u}\int_\Omega \frac{2}{d_1^2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x\big) - \tilde{u}\tilde{\psi}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x + \tilde{\lambda}_1\tilde{\phi}\int_\Omega \frac{1}{d_1}\mathrm{d}x = 0, \\ \tilde{\psi}\big(\displaystyle\int_\Omega \frac{r_2}{d_2}\mathrm{d}x - \tilde{u}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{2}{d_2^2}\mathrm{d}x\big) - \tilde{v}\tilde{\phi}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x + \tilde{\lambda}_1\tilde{\psi}\int_\Omega \frac{1}{d_2}\mathrm{d}x = 0. \end{cases} \tag{3.23}$$

Simplifying, we have

$$\begin{pmatrix} \tilde{\lambda}_1\displaystyle\int_\Omega \frac{1}{d_1}\mathrm{d}x - \tilde{u}\int_\Omega \frac{1}{d_1^2}\mathrm{d}x & -\tilde{u}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \\ -\tilde{v}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x & \tilde{\lambda}_1\displaystyle\int_\Omega \frac{1}{d_2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{1}{d_2^2}\mathrm{d}x \end{pmatrix}\begin{pmatrix} \tilde{\phi} \\ \tilde{\psi} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{3.24}$$

and therefore

$$\begin{vmatrix} \tilde{\lambda}_1\displaystyle\int_\Omega \frac{1}{d_1}\mathrm{d}x - \tilde{u}\int_\Omega \frac{1}{d_1^2}\mathrm{d}x & -\tilde{u}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \\ -\tilde{v}\displaystyle\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x & \tilde{\lambda}_1\displaystyle\int_\Omega \frac{1}{d_2}\mathrm{d}x - \tilde{v}\int_\Omega \frac{1}{d_2^2}\mathrm{d}x \end{vmatrix} = 0. \tag{3.25}$$

Notice by the Hölder inequality that

$$\begin{vmatrix} -\tilde{u}\int_\Omega \frac{1}{d_1^2}\mathrm{d}x & -\tilde{u}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \\ -\tilde{v}\int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x & -\tilde{v}\int_\Omega \frac{1}{d_2^2}\mathrm{d}x \end{vmatrix} \geq 0. \tag{3.26}$$

Thus $\tilde{\lambda}_1 > 0$, a contradiction. Therefore the proof for Part (i) of this theorem is finished.

Case 2. $a_5 < a_6, a_7 < a_8$.

Note that in this case $r_2 d_1 - \frac{\int_\Omega r_1 d_1^{-1}\mathrm{d}x}{\int_\Omega d_1^{-2}\mathrm{d}x} > 0$ at some point on $\bar{\Omega}$. Therefore, by Lemma 2.1 we get that $\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2}$ must achieves some positive value on $\bar{\Omega}$ for μ_1 sufficiently large. Similarly, $\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2}$ must achieves some positive value on $\bar{\Omega}$ for μ_1 sufficiently large.

Case 2.1. $a_1 > a_2, a_3 > a_4$.

In this case, by Lemma 2.1 one can get that

$$\lim_{\mu_1\to+\infty}\int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x = \int_\Omega \frac{r_2}{d_2}\mathrm{d}x - \int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \int_\Omega \frac{r_1}{d_1}\mathrm{d}x (\int_\Omega \frac{1}{d_1^2}\mathrm{d}x)^{-1} < 0, \tag{3.27}$$

and hence $\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1}) > 0$ for all μ_1 sufficiently large. By Proposition 2.3(ii), we obtain that for all μ_1 sufficiently large $\sigma_1(\mu_2 d_2, r_2 - \frac{\theta_1}{d_1}) < 0$ if $\mu_2 > \frac{1}{\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1})}$. Similarly, it follows that

$$\lim_{\mu_2\to+\infty}\int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x = \int_\Omega \frac{r_1}{d_1}\mathrm{d}x - \int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x \int_\Omega \frac{r_2}{d_2}\mathrm{d}x (\int_\Omega \frac{1}{d_2^2}\mathrm{d}x)^{-1} > 0, \tag{3.28}$$

and $\sigma_1(\mu_1 d_1, r_1 - \frac{\theta_2}{d_2}) > 0$ holds by Proposition 2.3(i) for any $\mu_1 > 0$ and μ_2 sufficiently large. Therefore, $(\theta_1, 0)$ is linearly stable and $(0, \theta_2)$ is linearly unstable for μ_1, μ_2 sufficiently large.

We now claim that there exists no coexistence steady state for all μ_1 and μ_2 sufficiently large. If not, then there exists a sequence $(\mu_1^{(n)}, \mu_2^{(n)})_{n=1}^\infty$ with $(\mu_1^{(n)}, \mu_2^{(n)}) \to (+\infty, +\infty)$ such that system (1.6) with $(\mu_1, \mu_2) = (\mu_1^{(n)}, \mu_2^{(n)})$ has a coexistence steady state (U_n, V_n), i.e.,

$$\begin{cases} \mu_1 d_1(r_1)\Delta U_n + U_n(r_1(x) - \frac{U_n}{d_1(r_1)} - \frac{V_n}{d_2(r_2)}) = 0, & \text{in } \Omega, \\ \mu_2 d_2(r_2)\Delta V_n + V_n(r_2(x) - \frac{U_n}{d_1(r_1)} - \frac{V_n}{d_2(r_2)}) = 0, & \text{in } \Omega, \\ \frac{\partial U_n}{\partial n} = \frac{\partial V_n}{\partial n} = 0, & \text{on } \partial\Omega. \end{cases} \tag{3.29}$$

By the maximum principle, $U_n > 0$, $V_n > 0$ hold on $\bar{\Omega}$. Furthermore, U_n and V_n are uniformly bounded in $C^{2,\alpha}(\bar{\Omega})$. We then can assume that $\lim_{n\to+\infty}(U_n, V_n) = (U_\infty, V_\infty)$ in $C^{2,\beta}(\bar{\Omega})$ for $\beta \in (0, \alpha)$, by passing to a subsequence of n if necessary. Indeed, dividing the

equation of U_n by $\mu_1^{(n)} d_1(r_1)$ and letting $n \to +\infty$ give rise to $\Delta U_\infty = 0$ in Ω, $\frac{\partial U_\infty}{\partial n} = 0$ on $\partial\Omega$, which imply that $U_\infty \geq 0$ is a constant. Similarly, $V_\infty \geq 0$ is also a constant. Setting $W_n = \frac{U_n}{\|U_n\|_\infty}$ and $Z_n = \frac{V_n}{\|V_n\|_\infty}$, by the same calculations above, we can get that $\lim_{n\to+\infty}(W_n, Z_n) = (1, 1)$. Next we show that this is impossible with the condition of this case.

Dividing the equation for U_n by $d_1\|U_n\|_\infty$ and the equation for V_n by $d_2\|V_n\|_\infty$ respectively, integrating over Ω and sending $n \to +\infty$, we obtain that

$$\begin{cases} U_\infty \int_\Omega \frac{1}{d_1^2} \mathrm{d}x + V_\infty \int_\Omega \frac{1}{d_1 d_2} \mathrm{d}x = \int_\Omega \frac{r_1}{d_1} \mathrm{d}x, \\ U_\infty \int_\Omega \frac{1}{d_1 d_2} \mathrm{d}x + V_\infty \int_\Omega \frac{1}{d_2^2} \mathrm{d}x = \int_\Omega \frac{r_2}{d_2} \mathrm{d}x. \end{cases} \tag{3.30}$$

However this is a contradiction since $\int_\Omega \frac{r_2}{d_2}\mathrm{d}x \int_\Omega \frac{1}{d_1^2}\mathrm{d}x < \int_\Omega \frac{r_1}{d_1}\mathrm{d}x \int_\Omega \frac{1}{d_1 d_2}\mathrm{d}x$. Hence we conclude that our claim is correct and the proof for Part (ii) of this theorem is finished.

Case 2.2. $a_1 < a_2, a_3 < a_4$.

Since this case can be deduced by the same methods as Case 2.1. above. Hence, Part (iii) of this theorem can be deduced similarly.

Next, we shall verify Part (iv)-(vii). Note from Proposition 3.1 that there exists a constant $\mu_i > 0$ $(i = 1, 2)$ such that

$$\theta_i = c_{r_i} + \frac{\rho_i + C_i}{\mu_i} + \frac{\gamma_i + E_i}{\mu_i^2} + O(\frac{1}{\mu_i^3}) \text{ for all } \mu_i > \mu_i. \tag{3.31}$$

Furthermore, there holds

$$\begin{cases} \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x = \frac{a_2 - a_1}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x} - \frac{1}{\mu_1}\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_1^2}) \text{ for all } \mu_1 > \mu_1, \\ \int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x = \frac{a_3 - a_4}{\int_\Omega \frac{1}{d_2^2}\mathrm{d}x} - \frac{1}{\mu_2}\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_2^2}) \text{ for all } \mu_2 > \mu_2. \end{cases} \tag{3.32}$$

Obviously, if $a_1 = a_2$ $(a_3 = a_4)$, then $a_3 > a_4$ $(a_2 > a_1)$.

Case 2.3. $a_1 = a_2, a_3 > a_4$.

This case indicates from (3.32) that

$$\begin{cases} \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\mathrm{d}x = -\frac{1}{\mu_1}\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_1^2}) \text{ for all } \mu_1 > \mu_1, \\ \int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x = \frac{a_3 - a_4}{\int_\Omega \frac{1}{d_2^2}\mathrm{d}x} - \frac{1}{\mu_2}\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_2^2}) \text{ for all } \mu_2 > \mu_2. \end{cases} \tag{3.33}$$

In this case, we have that $\int_\Omega (\frac{r_1}{d_1} - \frac{\theta_2}{d_1 d_2})\mathrm{d}x > 0$ for μ_2 large enough. By Proposition 2.3(i), one can have that $\sigma_1(\mu_1 d_1, r_1 - \frac{\theta_2}{d_2}) > 0$ and then $(0, \theta_2)$ is linearly unstable for all μ_2 sufficiently large and $\mu_1 > 0$.

Case 2.3.1. $\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x < 0$.

Clearly, we can obtain from the first equation of (3.33) that $(\theta_1, 0)$ is linearly unstable for all large μ_1 and $\mu_2 > 0$, if $\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2} \mathrm{d}x < 0$. Hence, Part (iv) is proved.

Case 2.3.2. $\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2} \mathrm{d}x > 0$, there holds $\int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2}) \mathrm{d}x < 0$ for large μ_1. Then we get that $\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1}) > 0$ for large μ_1. Moreover, from Proposition 2.3(ii), we have that for all large μ_1, $(\theta_1, 0)$ is linearly stable for $\mu_2 > \frac{1}{\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1})} =: \tilde{\mu}_2^*(\mu_1)$ and linearly unstable for all $\mu_2 < \tilde{\mu}_2^*(\mu_1)$.

Claim.

$$\mu_1 \lambda_1(d_2, r_2 - \frac{\theta_1}{d_1}) = c_{r_2} \int_\Omega \frac{\rho_1 + C_1}{d_1 d_2} \mathrm{d}x \Big/ \int_\Omega \rho_2 (\frac{r_2}{d_2} - \frac{c_{r_1}}{d_1 d_2}) \mathrm{d}x + o(1) \text{ as } \mu_1 \to +\infty. \tag{3.34}$$

To verify this claim, let φ_1 be the eigenfunction corresponding to $\lambda_1(d_2, r_2 - \frac{\theta_1}{d_1})$ such that $\|\varphi_1\|_\infty = 1$. Since it follows from $\lim_{\mu_1 \to +\infty} \int_\Omega (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2}) \mathrm{d}x = 0$ that $\lim_{\mu_1 \to +\infty} \lambda_1(d_2, r_2 - \frac{\theta_1}{d_1}) = 0$, due to standard elliptic regularity estimates, then we get a convergence(passing to a subsequence of μ_1 if necessary) of φ_1 to $\hat{\varphi}_1$ in $W^{2,p}(\Omega) \bigcap C^{1,\alpha}(\bar{\Omega})$ for some constant $\hat{\varphi}_1 \geq 0$ as $\mu_1 \to +\infty$. Since $\|\varphi_1\|_\infty = 1$, we have $\hat{\varphi}_1 = 1$. This implies that, as $\mu_1 \to +\infty$, $\varphi_1 = 1 + o(1)$.

Now denote $\lambda_1 := \lambda_1(d_2, r_2 - \frac{\theta_1}{d_1})$ for simplicity and rewrite $\varphi_1 = 1 + \frac{\lambda_1 \rho_2}{c_{r_2}} + \frac{\omega}{\mu_1^2}$. By direct calculations, ω is a solution satisfying

$$\begin{cases} \Delta \omega + \lambda_1 \cdot (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2}) \omega + R = 0 \text{ in } \Omega, \\ \frac{\partial \omega}{\partial n} = 0 \text{ on } \partial\Omega, \end{cases} \tag{3.35}$$

where

$$R = \frac{\lambda_1^2 \mu_1^2}{c_{r_2}} \rho_2 (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2}) + \lambda_1 \mu_1^2 (\frac{c_{r_2}}{d_2^2} - \frac{\theta_1}{d_1 d_2}). \tag{3.36}$$

Then $\int_\Omega R \varphi_1 \mathrm{d}x = 0$ holding follows directly by multiplying the equation for φ_1 (resp. ω) by ω (resp. φ_1) and integrating over Ω. Dividing both sides of $\int_\Omega R \varphi_1 \mathrm{d}x = 0$ by $\lambda_1 \mu_1$, combined with (3.31), (3.36) and $\varphi_1 = 1 + o(1)$, we have

$$\frac{\lambda_1 \mu_1}{c_{r_2}} \int_\Omega \rho_2 (1 + o(1)) (\frac{r_2}{d_2} - \frac{c_{r_1}}{d_1 d_2} + o(1)) \mathrm{d}x - \int_\Omega (\frac{\rho_1 + C_1}{d_1 d_2} + o(1))(1 + o(1)) \mathrm{d}x. \tag{3.37}$$

Therefore our claim follows by letting $\mu_1 \to +\infty$. Note that $\varphi_1 = 1 + o(1)$ can yield $\frac{\varphi_1 \int_\Omega \frac{r_2}{d_2} \mathrm{d}x}{\int_\Omega \frac{r_2}{d_2}\varphi_1 \mathrm{d}x} = 1 + o(1)$ as $\mu_1 \to +\infty$. We rewrite

$$\psi_1 := \frac{\varphi_1 \int_\Omega \frac{r_2}{d_2} \mathrm{d}x}{\int_\Omega \frac{r_2}{d_2}\varphi_1 \mathrm{d}x} = 1 + \frac{\lambda_1 \rho_2}{c_{r_2}} + \frac{\chi}{\mu_1^2}. \tag{3.38}$$

Owing to $\int_\Omega \rho_2 \mathrm{d}x = 0$, we get $\int_\Omega \chi \mathrm{d}x = 0$. Hence we have

$$\begin{cases} \Delta\chi + \lambda_1 \cdot (\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})\chi + R = 0 \text{ in } \Omega, \\ \frac{\partial \chi}{\partial n} = 0 \text{ on } \partial\Omega, \\ \int_\Omega \chi \mathrm{d}x = 0. \end{cases} \tag{3.39}$$

Combined with $\lim\limits_{\mu_1 \to +\infty} \lambda_1 = 0$, $\lim\limits_{\mu_1 \to +\infty} \theta_1 = c_{r_1}$ and $\lim\limits_{\mu_1 \to +\infty} (\Delta + \lambda_1(\frac{r_2}{d_2} - \frac{\theta_1}{d_1 d_2})) = \Delta$, since (3.34) yields the convergence of R, it is deduced that $\|\chi\|_\infty$ is bounded uniformly for all μ_1 large. Then there holds

$$\psi_1 := \frac{\varphi_1 \int_\Omega \frac{r_2}{d_2} \mathrm{d}x}{\int_\Omega \frac{r_2}{d_2}\varphi_1 \mathrm{d}x} = 1 + \frac{\lambda_1 \rho_2}{c_{r_2}} + O(\frac{1}{\mu_1^2}). \tag{3.40}$$

Similarly as before, we can show that $\int_\Omega R\psi_1 \mathrm{d}x = 0$. Using (3.31), (3.36) and dividing both sides of $\int_\Omega R\psi_1 \mathrm{d}x = 0$ by λ_1 yield

$$\begin{aligned} 0 = &\frac{\lambda_1 \mu_1^2}{c_{r_2}} \int_\Omega \rho_2 \big(\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2} - \frac{\rho_1 + C_1}{\mu_1 d_1 d_2} + O(\frac{1}{\mu_1^2})\big)\big(1 + \frac{\lambda_1 \rho_2}{c_{r_2}} + O(\frac{1}{\mu_1^2})\big)\mathrm{d}x \\ &+\mu_1 \int_\Omega \big(-\frac{\rho_1 + C_1}{d_1 d_2} - \frac{\gamma_1 + E_1}{\mu_1 d_1 d_2} + O(\frac{1}{\mu_1^2})\big)\big(1 + \frac{\lambda_1 \rho_2}{c_{r_2}} + O(\frac{1}{\mu_1^2})\big)\mathrm{d}x := I + II. \end{aligned} \tag{3.41}$$

By some further calculations, it follows that

$$\begin{cases} I = \lambda_1 \mu_1^2 \int_\Omega \frac{\rho_2}{c_{r_2}} (\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2})\mathrm{d}x + \lambda_1^2 \mu_1^2 \int_\Omega \frac{\rho_2^2}{c_{r_2}^2} (\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2})\mathrm{d}x \\ -\lambda_1 \mu_1 \int_\Omega \frac{\rho_2(\rho_1 + C_1)}{c_{r_2} d_1 d_2}\mathrm{d}x - \lambda_1^2 \mu_1 \int_\Omega \frac{\rho_2^2(\rho_1 + C_1)}{c_{r_2}^2 d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_1}), \\ II = -\mu_1 \int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x - \lambda_1 \mu_1 \int_\Omega \frac{\rho_2(\rho_1 + C_1)}{c_{r_2} d_1 d_2}\mathrm{d}x - \int_\Omega \frac{\gamma_1 + E_1}{d_1 d_2} + O(\frac{1}{\mu_1}). \end{cases} \tag{3.42}$$

Consequently,

$$\begin{aligned} &\mu_1 \Big(\int_\Omega \frac{\rho_1 + C_1}{d_1 d_2}\mathrm{d}x - \lambda_1 \mu_1 \int_\Omega \frac{\rho_2}{c_{r_2}} (\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2})\mathrm{d}x \Big) \\ &= \lambda_1^2 \mu_1^2 \int_\Omega \frac{\rho_2^2}{c_{r_2}^2} (\frac{r_2}{d_2} - \frac{c_{r_2}}{d_1 d_2})\mathrm{d}x - 2\lambda_1 \rho_2 \int_\Omega \frac{\rho_2(\rho_1 + C_1)}{c_{r_2} d_1 d_2}\mathrm{d}x - \int_\Omega \frac{\gamma_1 + E_1}{d_1 d_2}\mathrm{d}x + O(\frac{1}{\mu_1}). \end{aligned} \tag{3.43}$$

Dividing both sides of the above identity by $\lambda_1\mu_1^2\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x$, it follows from (3.34) that

$$\frac{1}{\lambda_1\mu_1}-\frac{\int_\Omega \frac{\rho_2}{c_{r_2}}(\frac{r_2}{d_2}-\frac{c_{r_2}}{d_1d_2})\mathrm{d}x}{\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x}=O(\frac{1}{\mu_1}). \tag{3.44}$$

Combining this identity with the right hand side of (3.43) yields

$$\frac{1}{\lambda_1}=\frac{\int_\Omega \frac{\rho_2}{c_{r_2}}(\frac{r_2}{d_2}-\frac{c_{r_2}}{d_1d_2})\mathrm{d}x}{\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x}\mu_1+B_1+O(\frac{1}{\mu_1}) \text{ for all } \mu_1>\mu_1, \tag{3.45}$$

where

$$\begin{aligned}B_1=&\frac{\int_\Omega \frac{\rho_2}{c_{r_2}}(\frac{r_2}{d_2}-\frac{c_{r_2}}{d_1d_2})\mathrm{d}x\int_\Omega \frac{\rho_2^2}{c_{r_2}^2}(\frac{r_2}{d_2}-\frac{c_{r_2}}{d_1d_2})\mathrm{d}x}{(\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x)^2}-\frac{2\int_\Omega \frac{\rho_2(\rho_1+C_1)}{c_{r_2}d_1d_2}\mathrm{d}x}{\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x}\\&-\frac{\int_\Omega \frac{\rho_2}{c_{r_2}}(\frac{r_2}{d_2}-\frac{c_{r_2}}{d_1d_2})\mathrm{d}x\int_\Omega \frac{\gamma_1+E_1}{d_1d_2}\mathrm{d}x}{(\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x)^2}.\end{aligned} \tag{3.46}$$

Therefore we complete the proof of Part (v).

Case 2.4. $a_1<a_2, a_3=a_4$.

This case indicates from (3.32) that

$$\begin{cases}\int_\Omega(\frac{r_2}{d_2}-\frac{\theta_1}{d_1d_2})\mathrm{d}x=\frac{a_2-a_1}{\int_\Omega \frac{1}{d_1^2}\mathrm{d}x}-\frac{1}{\mu_1}\int_\Omega \frac{\rho_1+C_1}{d_1d_2}\mathrm{d}x+O(\frac{1}{\mu_1^2}) \text{ for all } \mu_1>\mu_1,\\ \int_\Omega(\frac{r_1}{d_1}-\frac{\theta_2}{d_1d_2})\mathrm{d}x=-\frac{1}{\mu_2}\int_\Omega \frac{\rho_2+C_2}{d_1d_2}\mathrm{d}x+O(\frac{1}{\mu_2^2}) \text{ for all } \mu_2>\mu_2.\end{cases} \tag{3.47}$$

Case 2.4.1. $\int_\Omega \frac{\rho_2+C_2}{d_1d_2}\mathrm{d}x<0$.

By similar arguments as in the proof for Case 2.3.1, both $(\theta_1,0)$ and $(0,\theta_2)$ are linearly unstable for all large μ_1,μ_2. Hence, Part (vi) is proved.

Case 2.4.2. $\int_\Omega \frac{\rho_2+C_2}{d_1d_2}\mathrm{d}x>0$.

In this case, similar to Case 2.3.2, we get that $\lambda_1(d_1,r_1-\frac{\theta_2}{d_2})>0$ for large μ_2. Moreover for all large μ_2, $(0,\theta_2)$ is linearly stable for $\mu_1>\frac{1}{\lambda_1(d_1,r_1-\frac{\theta_2}{d_2})}=:\tilde{\mu}_1^*(\mu_2)$ and linearly unstable for all $\mu_1<\tilde{\mu}_1^*(\mu_2)$. Here $\frac{1}{\lambda_1(d_1,r_1-\frac{\theta_2}{d_2})}$ satisfies the asymptotic expansion

$$\frac{1}{\lambda_1(d_1,r_1-\frac{\theta_2}{d_2})}=\frac{\int_\Omega \frac{\rho_1}{c_{r_1}}(\frac{r_1}{d_1}-\frac{c_{r_1}}{d_1d_2})\mathrm{d}x}{\int_\Omega \frac{\rho_2+C_2}{d_1d_2}\mathrm{d}x}\mu_2+B_2+O(\frac{1}{\mu_2}) \text{ for all large } \mu_2, \tag{3.48}$$

where

$$B_2 = \frac{\int_\Omega \frac{\rho_1}{c_{r_1}}(\frac{r_1}{d_1} - \frac{c_{r_1}}{d_1 d_2})\mathrm{d}x \int_\Omega \frac{\rho_1^2}{c_{r_1}^2}(\frac{r_1}{d_1} - \frac{c_{r_1}}{d_1 d_2})\mathrm{d}x}{(\int_\Omega \frac{\rho_2 + C_2}{d_1 d_2}\mathrm{d}x)^2} - \frac{2\int_\Omega \frac{\rho_1(\rho_2+C_2)}{c_{r_1} d_1 d_2}\mathrm{d}x}{\int_\Omega \frac{\rho_2+C_2}{d_1 d_2}\mathrm{d}x} - \frac{\int_\Omega \frac{\rho_1}{c_{r_1}}(\frac{r_1}{d_1} - \frac{c_{r_1}}{d_1 d_2})\mathrm{d}x \int_\Omega \frac{\gamma_2+E_2}{d_1 d_2}\mathrm{d}x}{(\int_\Omega \frac{\rho_2+C_2}{d_1 d_2}\mathrm{d}x)^2}. \tag{3.49}$$

Therefore we complete the proof of Part (vii).

The proof of this theorem is therefore completed. □

Declarations

Competing Interests The author declares no competing interests.

References

1. Brown, K.J., Lin, S.S.: On the existence of positive eigenfunctions for an eigenvalue problem with indefinite weight function. J. Math. Anal. Appl. **75**(1), 112–120 (1980)
2. Cantrell, R.S., Cosner, C.: Spatial Ecology via Reaction-Diffusion Equations. Series in Mathematical and Computational Biology. Wiley, Chichester (2003)
3. de Mottoni, P.: Qualitative analysis for some quasi-linear parabolic systems. Inst. Math. Polich Acad. Sci. Zam. **190**, 11–70 (1979)
4. Dockery, J., Hutson, V., Mischaikow, K., Pernarowski, M.: The evolution of slow dispersal rates: a reaction-diffusion model. J. Math. Biol. **37**, 61–83 (1998)
5. He, X., Ni, W.-M.: Global dynamics of the Lotka-Volterra competition-diffusion system: diffusion and spatial heterogeneity I. Commun. Pure Appl. Math. **69**, 981–1014 (2016)
6. He, X., Ni, W.-M.: Global dynamics of the Lotka-Volterra competition-diffusion system with equal amount of total resources, II. Calc. Var. Partial Differ. Equ. **55**, 25 (2016)
7. He, X., Ni, W.-M.: Global dynamics of the Lotka-Volterra competition-diffusion system with equal amount of total resources, III. Calc. Var. Partial Differ. Equ. **56**, 132 (2017)
8. Hess, P., Senn, S.: Another approach to elliptic eigenvalue problems with respect to indefinite weight functions. In: Nonlinear Analysis and Optimization, Bologna, 1982. Lecture Notes in Mathematics, vol. 1107, pp. 106–114. Springer, Berlin (1984)
9. Krein, M.G., Rutman, M.A.: Linear operators leaving invariant a cone in a Banach space. Usp. Mat. Nauk **3**, 3–95 (1948)
10. Lam, K.-Y., Ni, W.-M.: Uniqueness and complete dynamics in the heterogeneous competition-diffusion systems. SIAM J. Appl. Math. **72**, 1695–1712 (2012)
11. Lou, Y.: On the effects of migration and spatial heterogeneity on single and multiple species. J. Differ. Equ. **223**, 400–426 (2006)
12. Lou, Y.: Some challenging mathematical problems in evolution of dispersal and population dynamics. In: Tutorials in Mathematical Biosciences. IV. Lecture Notes in Math., vol. 1922, Math. Biosci. Subser., pp. 171–205. Springer, Berlin (2008)
13. Lou, Y., Yanagida, E.: Minimization of the principal eigenvalue for an elliptic boundary value problem with indefinite weight, and applications to population dynamics. Jpn. J. Ind. Appl. Math. **23**(3), 275–292 (2006)
14. Ni, W.-M.: The Mathematics of Diffusion. CBMS-NSF Regional Conference Series in Applied Mathematics, vol. 82. SIAM, Philadelphia (2011)
15. Qin, W.Y., Zhou, P.: A review on the dynamics of two species competitive ode and parabolic systems. J. Appl. Anal. Comput. **12**, 2075–2109 (2022)
16. Smith, H.: Monotone Dynamical System. An Introduction to the Theory of Competitive and Cooperative Systems. Math. Surveys Monogr., vol. 41. Am. Math. Soc., Providence (1995)
17. Tang, D., Wang, Z.-A.: Population dynamics with resource dependent dispersal single and two species models. J. Math. Biol. **86**, 23 (2023)

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